

TitleChain

Turning a Tamil Nadu Encumbrance Certificate into a verified ownership-and-encumbrance graph – every live encumbrance, chain break, and search-window gap flagged deterministically, and traceable to its source region on the page.

VERSION	DATE	OWNER	EVENT
1.0	26 July 2026	bsraavan@gmail.com	Sarvam Epoch Buildathon

Status: **Idea locked**. Hardest dependency empirically de-risked on real, public Encumbrance Certificates – see \$0.2.

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0 Pre-PRD: Rubric Validation

This section exists because the rubric — not a generic product standard — is the objective function. Everything downstream is designed against it.

0.1 Declared job (fixed, do not drift)

THE JOB

Given a Tamil Nadu Encumbrance Certificate, produce a title-scrutiny-ready chain report: the property's ownership/encumbrance graph, with every live encumbrance, chain break, and search-window gap identified deterministically and traceable to its source region on the page.

0.2 Evidence already in hand (run 26 Jul 2026, pre-event validation)

INPUT	PROVENANCE	PAGES	ANCHORS INTACT
EC-A	Pollachi SRO EC, public AICTE filing	3	23/24 = 96%
EC-A'	same, re-rasterised 120 dpi grayscale JPEG	3	24/24 = 100%
EC-B	Pollachi SRO EC, public RERA filing	6	22/22 = 100%
Total	real documents, not hand-authored	12	69/70 = 98.6%

Ground truth was read directly from the source PDFs by a human/model reading the original, **not** from pipeline output. Checker: `check_ec_integrity.py`. Runtime 12 pages ≈ 60 s. Cost ≈ ₹6.

Two findings that change the design

1. The undocumented per-page JSON contains provenance. Every block carries `block_id`, `coordinates{x1,y1,x2,y2}`, `layout_tag`, `confidence`, `reading_order`, and `text` (tables as HTML). Bounding-box provenance and confidence-ranked review move from "stretch goal" to **verified Explicit capability**. Image space is 3509×2480 (300 dpi).
2. confidence is **layout confidence**, not **transcription confidence**. The single lost anchor (entry 2's three dates) was dropped from a table block scoring **0.96**. Thresholding on the API's confidence would **not** have caught it. Therefore the uncertainty net must be **our own schema-completeness validator** (§10.6) — proven necessary by real data, not assumed, and a moat component, not a vendor feature.

Observed confidence distribution across 48 blocks / 12 pages:

LAYOUT_TAG	N	MIN	MEDIAN	MAX
table	15	0.64	0.95	0.98
paragraph	14	0.29	0.65	0.86
page-number	11	0.26	0.34	0.70
header	3	0.38	0.41	0.46
footer	3	0.28	0.38	0.60
footnote	2	0.75	0.80	0.80

Tables — the only blocks we care about — are reliably high-confidence. Decorative regions are low. The signal is discriminative for *region triage*, useless for *cell omission*.

0.3 Honest level assessment

PARAMETER	TODAY	TARGET	WHY NOT L5 YET	CONCRETE FIX BEFORE/DURING BUILD
Job-to-be-done	L3	L5	Stage 1 verified on 3 inputs; stage 2 (HTML→typed rows) untested . L5 needs 85%+ on ≥3 repeated cases, no builder rescue.	Run stage-2 dry run tonight (§14.0). Source 3 more ECs from different SROs. Demo on 3 unseen ECs live.
Memory & Context	L2	L4→L5	Nothing persists yet. L5 demands current task + case history + business rules across sessions/handoffs, with corrections propagating, staleness visible, permissions intact.	SQLite case store from hour 2. Correction table that re-derives the graph. Rulebook versioning. EC staleness clock. Advocate-scoped case ownership.

PARAMETER	TODAY	TARGET	WHY NOT L5 YET	CONCRETE FIX BEFORE/DURING BUILD
Creativity	L4	L5	The graph reframing + PR-pointer edges + window-insufficiency detection are several reinforcing choices. L5 needs the "I didn't know you could solve it that way" beat.	Lead the demo with search-window insufficiency → pre-filled next EC order . The non-obvious unlock.
Impact	L3	L4→L5	Numbers are currently order-of-magnitude assumptions, not defended baselines. Rubric: market size alone is not impact.	Build the arithmetic explicitly with stated assumptions and sensitivity (\$4). Anchor on one defensible number: hours per scrutiny.
Delight	L2	L4→L5	Not built. The friction moment is "did I miss something, and is this EC even sufficient?"	Answer it: the insufficiency verdict + exact next action (SRO, village, survey, date range) pre-filled. Stay with the user past the immediate result.
Document Intelligence	L4	L5	Real Tamil ECs with dense multi-page tables pass at 98.6%, and uncertain regions are inspectable against source. L5 wants the hardest material + competing readings visible.	Add: true phone-photo input class (\$14.0), structural-gap flags rendered as competing/missing readings with source crops.

WEAKEST LINKS, RANKED

(1) stage-2 extraction unverified · (2) both real ECs are from the *same SRO* (Pollachi) — formatting luck is untested · (3) the "degraded scan" was synthetic re-rasterisation, not a real phone photo with skew/shadow/glare · (4) Memory is entirely unbuilt · (5) Impact baseline is assumption-heavy.

0.4 Sarvam capability selection

Chosen: Document Intelligence. Scored deeply, and it is the only capability the declared job actually requires.

Why it is the strongest choice. The job's hard input is a dense, multi-page, Tamil-script government table where names, deed types, survey numbers and previous-document pointers must survive **as structured rows with source coordinates**. That is precisely what Sarvam Vision returns and precisely where frontier VLMs degrade. Swap-out test: replace Sarvam with a GPT/Gemini-class VLM and the Tamil table flattens, reading order scrambles, and — decisively — you lose the per-block coordinates that make every finding traceable. The degradation is severe and visible on the hardest case. This is a genuine structural advantage, not a preference.

Why Voice is unnecessary and harmful. This is a desk workflow. A lawyer reviewing title does not want to talk; they want a document they can attach to an opinion. Adding voice would (a) score zero — additional capabilities add no points, (b) land us on the most crowded ground at the event (the Sarvam cookbook ships collections, government-scheme, tutor and loan-advisory voice agents; reproducing that shape is an explicit creativity handicap), and (c) consume hours on turn-taking and barge-in quality that the rubric would score under a parameter we did not declare.

Why Dubbing is unnecessary and impossible. There is no dubbing API — it is a Sarvam Studio dashboard workflow. It cannot be a programmatic dependency, and nothing about title scrutiny involves media adaptation.

Additional capabilities used, and only because the product needs them: sarvam-105b (row typing into strict JSON — the alternative is brittle regex over Tamil HTML) and the Transliteration API (cross-script party-name clustering — genuinely load-bearing for chain-break detection). Neither is decorative. No TTS, no STT, no dubbing.

1 Executive Summary

ONE SENTENCE

TitleChain turns a Tamil Nadu Encumbrance Certificate into a verified ownership-and-encumbrance graph, flagging every live encumbrance, chain break, and – critically – whether the certificate even covers enough history to support the title search it was ordered for.

Why it should exist

Every property sale and home loan in Tamil Nadu requires a title search over 13–30 years. The EC is the primary instrument for that search: a dense Tamil table, one row per registered transaction. Today an advocate reads it row by row with a highlighter and assembles the ownership chain in their head or on paper. Three failure modes recur, and all three are mechanical rather than intellectual: a cancelled instrument is read as a live encumbrance (or worse, a live one is missed); a parent-document pointer leads outside the certificate's search window and nobody notices the chain is unverified; and a name spelled two ways across entries hides a break in the chain. The cost of each miss is borne by the buyer, years later, in litigation.

Why now

The EC already contains explicit machine-readable chain edges — the *PR Number* (முந்தைய ஆவண எண், "previous document number") field — that nobody assembles into a graph. Until now nothing could read the Tamil table reliably enough to trust those pointers. As of this month, Sarvam Vision does, at 98.6% field fidelity on real certificates, with coordinates for every block. The raw material for the graph has existed for years; the reader arrived this year.

Why this is an ideal Buildathon project

The hardest dependency is already empirically de-risked on real, publicly obtained documents. The demo input is genuinely hard (dense Tamil government table) and genuinely real. The output is a usable artifact a professional would attach to a file. The winning insight — an EC can be *insufficient for its own purpose*, and we can prove it and pre-fill the fix — is invisible from the idea card and unlikely to occur to another team.

98.6%

FIELD-ANCHOR FIDELITY, REAL ECS

<₹5

COST PER CERTIFICATE PROCESSED

~40%

REDUCTION IN SCRUTINY EFFORT

2 Idea Lock

Fixed for the duration of the build. Changes require deleting a feature, not adding one.

One-sentence product	Upload a TN Encumbrance Certificate; get the property's ownership chain as a verified graph — every link traceable to its source region, every live encumbrance, chain break and search-window gap flagged deterministically.
Target user	Property advocate / bank-empanelled lawyer performing pre-sale or pre-mortgage title scrutiny in Tamil Nadu.
Job completed	A title-scrutiny-ready chain report, produced end to end from a raw EC, with findings a lawyer can sign off on or contest against the source.
Hard input	Multi-page Tamil-script government EC with dense nested tables, mixed Tamil/English, bilingual column headers, Tamil party names, and cross-referencing remarks. Scanned or digital.
Final artifact / state change	A persisted case file containing: typed transaction entries, the ownership graph, a ranked findings list, and an exportable scrutiny report. Corrections persist and re-derive the graph.
Sarvam capability	Document Intelligence (declared, scored)
Exact Sarvam APIs	<code>document_intelligence create/upload/start/status/download (Sarvam Vision, output_format=md, language=ta-IN) · chat.completions sarvam-105b with response_format=json_schema strict · text.transliterate ta-IN→en-IN</code>
Additional capability justification	105B: converts digitised Tamil table HTML into typed rows — regex over Tamil HTML is brittle and unmaintainable. Transliteration: normalises party names across scripts so chain-break detection works. Nothing else.
Supported language / input subset	Tamil + English TN ECs, post-1987 computerised registration era, ≤10 pages per job (chunked beyond).
Team advantage	We already ran the kill test on real certificates before the event and know exactly what the API returns — including that its confidence score cannot catch cell omissions, which most teams will discover too late or never.
Creativity thesis	An EC is not a document to read — it is a graph to reconstruct. The PR Number field is an explicit parent pointer that nobody follows; following it reveals that most ECs cannot support the title search they were ordered for , and we can compute the exact replacement certificate to order.
Delight thesis	The lawyer's real fear is not "what does this say" but "what did I miss." We answer with a deterministic, evidence-cited verdict — including honest admissions of what this certificate <i>cannot</i> tell them — and a pre-filled next action instead of a shrug.
Demo proof	On a real, unmodified EC: a Lease deed that looks like a live encumbrance is shown to have been cancelled by a later deed; simultaneously, five parent documents from 2005–2011 are shown to fall outside the certificate's 2018–2023 window, so the chain is unverified — with the exact earlier EC to order, pre-filled.
Non-goals	Handwritten pre-1987 records · Modi/Kaithi/Grantha scripts · other states · voice/TTS/dubbing · patta/chitta/FMB integration · legal opinions (we produce evidence and flags; the advocate signs) · TNREGINET scraping/automation

3 Problem Statement

3.1 The current workflow

1. Buyer/bank instructs an advocate to conduct title scrutiny, statutorily over 13 years, in practice often 30.
2. The advocate obtains an EC from TNREGINET or the sub-registrar's office for a stated search period and survey number(s).
3. The EC arrives as a PDF: a header block (SRO, village, survey details, search period) and then a table, one numbered entry per registered transaction. Each entry carries: document number & year, three dates (execution / presentation / registration), nature of deed, executant(s), claimant(s), volume & page, consideration value, market value, **PR Number (previous document numbers)**, document remarks, and one or more property schedules with type, extent, survey numbers, door number and boundaries.
4. The advocate reads every entry, mentally assembles who transferred what to whom in what order, notes mortgages and whether they were discharged, and cross-checks survey numbers and extents for drift.
5. They write a scrutiny report / title opinion.

3.2 Why the current process fails

These are mechanical failure modes, which is exactly why software should own them:

- **The cancellation trap.** A deed and the instrument cancelling it appear as separate numbered entries, often pages apart, linked only by a sentence in a remarks cell. Read linearly, a cancelled lease looks like a live encumbrance. Read carelessly in the other direction, a live mortgage looks resolved.
- **The silent window gap.** PR Numbers routinely point to documents registered *before* the EC's search period. The chain therefore continues outside the certificate in hand. Nothing on the document announces this. The advocate must notice, unprompted, that the evidence is insufficient — and it is easy not to.
- **Name-variance breaks.** Tamil party names appear with differing transliterations, honorifics, initials and parenthetical role markers (e.g. (முத.), (முத.)) across entries. A break between claimant *n* and executant *n+1* can be real or an artefact of spelling; distinguishing the two by eye is unreliable.
- **Survey and extent drift.** Subdivisions renumber survey plots. Extents change across entries for legitimate (partition) and illegitimate reasons. Tracking this across a 30-year EC by hand is drudgery with high stakes.
- **It does not scale.** The work is linear in the number of entries and is performed by the most expensive person in the chain.

3.3 Who, how often, what it costs

- **Who:** property advocates and bank-empanelled lawyers in TN; secondarily the buyer and the lending bank who inherit the risk.
- **How often:** every property transaction requiring a loan or diligence — a continuous, non-seasonal workload for the profession.
- **Cost today:** hours of senior professional time per file (assumption, \$4.1), and on failure, a title defect discovered years later during resale or foreclosure — the most expensive possible moment.
- **Why now:** the reading problem is newly solved (\$0.2), and the chain edges were always in the data.

3.4 Existing alternatives and why they fail

ALTERNATIVE	WHAT IT DOES	WHY IT FAILS THIS JOB
Manual reading (status quo)	Advocate reads EC	Slow, linear, and the three failure modes above are attention failures under time pressure
Record-retrieval platforms (e.g. Landeed and similar)	Fetch land records/ECs quickly across states	Solves <i>access</i> , not <i>interpretation</i> . You still receive an unread table.
Generic LLM (upload the PDF to ChatGPT/Gemini)	Summarises the document	Tamil table fidelity degrades; more importantly it will <i>narrate</i> a conclusion with no deterministic rulebook, no provenance, and no ability to say "this certificate is insufficient." A confident wrong answer on an encumbrance is worse than no answer.
Generic OCR / IDP suites	Extract text and fields	Produce a text dump or flat fields. No chain, no graph, no domain rules, no Indic-script table fidelity.
Bank legal-vetting vendors	Human-powered scrutiny at scale	Labour arbitrage, not tooling. Same failure modes, outsourced.

4 Business Value

Constructed as a defensible argument, with assumptions labelled. Per the rubric: market size alone is not impact.

4.1 Assumption register

Every number below is an assumption to be verified before external use. They are stated so they can be attacked.

#	ASSUMPTION	VALUE USED	CONFIDENCE	HOW TO VERIFY
A1	Advocate time per title scrutiny (EC reading + chain assembly)	3–8 h, midpoint 5 h	Medium	Interview 5 TN property advocates
A2	Share of that time that is EC reading/chain assembly (vs. drafting, client comms)	50–60%	Medium	Same interviews
A3	Professional fee per scrutiny	₹3,000–15,000	Medium	Published empanelment schedules; practitioner interviews
A4	TN document registrations per year	order 10^6 – 10^7	Low — order of magnitude only	IGRS TN annual administration report
A5	Share of transactions requiring formal scrutiny (loan/diligence-backed)	30–50%	Low	Bank empanelment data
A6	Reduction in EC-reading time with TitleChain	60–80%	Medium — grounded in 98.6% field fidelity + deterministic rules	Timed A/B with practising advocates

4.2 The value calculation

Take one advocate, one file. Under A1/A2, EC reading and chain assembly consume ≈ 2.5 – 3 h of a 5 h scrutiny. Under A6, TitleChain reduces this to ≈ 30 – 60 min of *review* — the advocate is checking flagged findings against source crops rather than reading a table.

THE NUMBER THAT MATTERS

Time saved per file: ~ 2 hours, i.e. $\sim 40\%$ of total scrutiny effort. That clears the rubric's L5 bar ($>30\%$ movement on an operating metric) on the metric that matters to the payer, and it survives challenge even at the pessimistic end of A6 (60% reduction $\rightarrow \sim 1.5$ h $\rightarrow \sim 30\%$).

Second-order value, harder to quantify but larger: the window-insufficiency check converts an invisible risk into a caught defect. An advocate who unknowingly opines on a chain whose parents lie outside the search window has produced an unsound opinion. We do not know the base rate of this (verify: A7), but its cost is not the ₹5,000 fee — it is the litigated value of the property. Both ECs we tested exhibit PR pointers outside their search window. Two out of two is not a base rate, but it is a signal that the pattern is common rather than exotic.

4.3 Why customers pay

The buyer is the advocate or the firm, and the purchase is straightforward: a tool that returns ~ 2 hours per file and reduces professional-liability exposure, priced far below the fee for a single file. Per-EC processing cost is **under ₹5** (₹0.5/page $\times$ 3–6 pages, plus negligible LLM tokens). Even at a per-file price of ₹200–500 the gross margin is $\sim 98\%$ and the customer's ROI is immediate and obvious. Banks are the second payer: they already fund this work through empanelment and bear the residual risk.

4.4 TAM — stated last, deliberately

Under A4/A5, the annual volume of TN transactions requiring scrutiny is plausibly in the low millions; at ₹200–500 per file that is a market in the $\text{₹}10^2$ – 10^3 crore range for Tamil Nadu alone, before expansion. **This number is the least reliable thing in this document and the least important.** The urgency argument in §4.2 is what should be defended; TAM is context.

4.5 Defensibility

The moat is deliberately placed in our layer, never in Sarvam's model quality — which any competitor can buy tomorrow.

1. **The rulebook.** Encoded TN registration practice: deed-nature taxonomy, discharge semantics, cancellation cross-reference grammar, the meaning of (முத.) / (முத.) role markers, subdivision conventions. This is institutional knowledge, acquired from practitioners, not derivable from the document.

2. **Correction memory.** Every advocate correction (a merged name cluster, a re-typed deed nature, a rejected finding) is stored, attributed and replayed. The system's accuracy on *this firm's* districts and vocabulary compounds with use. A competitor starts at zero corrections.
3. **The name graph.** Accumulated party-name clusters across cases become a proprietary entity graph — the same trust, the same builder, the same family appearing across files. This is a genuine network effect *within a firm* and, if pooled with consent, across firms.
4. **Workflow lock-in.** Once the case file is the working surface for scrutiny, the report, the corrections and the audit trail live here.

4.6 Network effects

Weak-to-moderate and honestly bounded: the name graph and rulebook improve with volume, and improvements are shared across all users of the same district profile. This is a data network effect, not a marketplace one — it compounds but does not create winner-take-all dynamics.

4.7 Expansion path

EC chain reconstruction (TN) → sale-deed cross-verification against the EC → patta/chitta reconciliation → other computerised states with equivalent instruments (KA, TS, AP each have analogous certificates and the same PR-pointer logic) → bank-side portfolio scanning (re-scrutinise an existing loan book for defective titles). Each step reuses the graph and the rulebook.

4.8 Long-term vision

A verified, machine-readable title graph for Indian property, assembled bottom-up from the documents transactions already generate — with provenance to the source region of the source certificate for every edge. The registry that the registry does not provide.

4.9 Biggest risks and assumptions

RISK	SEVERITY	MITIGATION
A1/A3 wrong — advocates value their time differently than assumed	High for pricing, low for product	Interviews before pricing; the time-saving claim holds even at pessimistic A6
Formatting varies materially across SROs/districts	High — currently untested	Source ECs from ≥3 districts tonight (\$14.0)
Advocates distrust automated findings	Medium	Every finding is deterministic and cites a source crop; the product never opines, it evidences
Regulatory/liability exposure for a wrong flag	Medium	Positioned as an evidence tool with mandatory human sign-off; no legal opinion is generated
TNREGINET changes EC layout	Low-medium	Layout-agnostic extraction (LLM over digitised HTML, not fixed coordinates)

5 User Personas

Primary – "Advocate Meena", property advocate, Coimbatore

- **Practice:** 15 years; bank empanelment with two nationalised banks; 8–15 scrutiny files a month.
- **Goals:** clear the file accurately, protect her name on the opinion, keep turnaround inside the bank's SLA.
- **Pain points:** the tedium is the risk — attention fades on entry 40 of a 30-year EC, and that is exactly where the cancelled-mortgage entry sits. She has no systematic way to know whether an EC's window actually covers the chain.
- **Current workaround:** printouts, highlighters, a handwritten chain on the file cover, and re-reading when something feels off.
- **Why she switches:** two hours back per file, and a written record showing she checked what she checked.

Secondary – "Credit Manager Ravi", nationalised bank branch, Salem

- **Goals:** disburse fast without booking a bad title; defensible file if the loan sours.
- **Pain points:** cannot evaluate the quality of a scrutiny report he receives; discovers defects only at foreclosure.
- **Current workaround:** trusts the empanelled advocate entirely.
- **Why he adopts:** a structured, evidence-linked chain report attached to the loan file — and an explicit statement of what the search did *not* cover.

Tertiary – "Buyer Karthik" (beneficiary, not a user in v1)

Bears the entire downside of a missed encumbrance and has no ability to verify. Out of scope for the build; the reason the product matters.

6 User Journey

TODAY

Order EC

↓

Receive dense Tamil PDF

↓

Read 40 entries linearly

- ✗ Miss the cancellation cross-reference
- ✗ Never notice PR pointers precede the search window
- ✗ Mis-read name variance as a chain break (or miss a real one)
- ✗ Track survey/extent drift by memory

↓

Write opinion under time pressure

↓

Residual risk transfers silently to the buyer

WITH TITLECHAIN

Upload EC

↓

Sarvam Vision digitises (tables + coordinates + reading order)

↓

105B types each entry into a strict schema

↓

Transliteration clusters party names across scripts

↓

OUR deterministic rulebook builds the graph and runs 10 checks

↓

CHAIN GRAPH & FINDINGS (RANKED, EVIDENCE-CITED)

o 2520/2019 Lease ↳ cancelled by —▶ 8756/2020 Cancel

⚠ WINDOW INSUFFICIENT — 5 parent docs (2005–2011) outside the 2018–2023 search period

✓ Lease extinguished — not a live encumbrance

📄 Entry 2: registration date missing → review source

↓

Every finding clicks to its bounding-box crop on the source page

↓

Advocate corrects one name → graph re-derives → correction persists

↓

NEXT ACTION – PRE-FILLED

Order EC · Pollachi SRO · Puliampatti · survey 95/2, 100/3A, 113/1B, 116/A1, 116/B1 · 01-Jan-1993 → 31-Dec-2017



Export scrutiny report → attach to file

WHERE VALUE IS CREATED

Not at extraction (that is table stakes), but at the two moments a human reliably fails — the cross-referenced cancellation, and the unannounced insufficiency of the evidence itself.

7 Core MVP

Ruthlessly scoped. Every feature names the rubric parameter it serves and its status. **If we are behind, cut from the bottom.**

#	FEATURE	WHY IT EXISTS	RUBRIC PARAMETER	STATUS
F1	EC upload → Sarvam Vision digitisation (chunked >10 pages)	The hard input must be read	Document Intelligence	MANDATORY
F2	Entry typing into strict JSON schema (105B)	Rows must become records before they can become a graph	JTBD	MANDATORY
F3	Graph assembly from PR pointers + remark cross-references	The core reframing	Creativity	MANDATORY
F4	Deterministic rulebook — 10 checks (\$10.5)	The findings a lawyer acts on; legally sensitive logic must not be model output	JTBD	MANDATORY
F5	Window-insufficiency detection + pre-filled next EC order	The non-obvious unlock and the delight beat	Delight	MANDATORY
F6	Source-crop provenance (bbox → cropped image per finding)	Traceability; makes review precise	Document Intelligence	MANDATORY
F7	Schema-completeness validator → uncertainty flags	Catches cell omissions that confidence cannot (\$0.2)	Document Intelligence	MANDATORY
F8	Persisted case file (SQLite) + correction propagation	Memory L4; corrections re-derive the graph	Memory & Context	MANDATORY
F9	Party-name clustering via transliteration	Chain-break detection is unreliable without it	JTBD	MANDATORY
F10	Exportable scrutiny report	The final usable artifact	JTBD	MANDATORY
F11	Advocate login + case ownership scoping	Memory L5 requires permissions intact	Memory & Context	HIGH-VALUE OPTIONAL
F12	EC staleness clock (issue date vs today)	Stale vs current information distinguishable — explicit L5 wording	Memory & Context	HIGH-VALUE OPTIONAL
F13	Double-pass extraction disagreement flags	Second uncertainty signal	Document Intelligence	OPTIONAL – CUT FIRST
F14	Multi-EC merge into one property chain	Expansion demo	—	PARKING LOT
F15	Sale-deed cross-verification	Expansion demo	—	PARKING LOT

EXPLICITLY CUT

Any voice interface, any dashboard analytics, any avatar/animation, multi-state support, TNREGINET integration, mobile app, real-time collaboration.

8 Functional Requirements

FR-1 Ingest

Accept PDF/JPEG/PNG ≤50 MB. Split >10-page PDFs into ≤10-page jobs client-side; stitch results preserving global page numbering. Reject unsupported types with a specific message.

FR-2 Digitise

For each chunk: `create_job(language="ta-IN", output_format="md")` → `upload_file` → `start` → `poll` → `download_output`. Persist both `document.md` and every `metadata/page_NNN.json` verbatim. Never discard the raw response — it is the provenance record.

FR-3 Table isolation

From page metadata, select blocks with `layout_tag == "table"`, ordered by `reading_order`. Retain `block_id`, coordinates, confidence, `page_num` alongside the HTML.

FR-4 Entry typing

For each EC (or page group), call `sarvam-105b` with `response_format={"type": "json_schema", ...}` strict, supplying the table HTML and the schema in §10.4. The model transcribes and structures; it must not infer, summarise, or resolve conflicts. Unreadable field → explicit `null`, never a guess.

FR-5 Provenance binding

Every typed entry stores the `page_num` and `block_id` it came from, so any field can be rendered as a cropped region of the source page image.

FR-6 Name normalisation

For each party name, call `text.transliterate ta-IN→en-IN`; store native and roman forms; strip honorifics and parenthetical role markers into a separate `role_marker` field; cluster by normalised-token similarity within a case.

FR-7 Graph assembly

Nodes = entries. Edges: `PR_PARENT` (from PR Number), `CANCELS / CANCELLED_BY` (parsed from remarks), `SUCCESSION` (claimant of entry *n* clusters with executant of entry *m*, *m* later). Edges to documents absent from this EC become **dangling nodes**, explicitly typed as unexamined.

FR-8 Rulebook execution

Run all ten checks (§10.5) as pure deterministic functions over the typed graph. Each emits `{rule_id, severity, message, evidence: [entry_ids], source_crops: [...]}`. **No rule may call a model.**

FR-9 Findings presentation

Rank by severity (blocking → material → informational). Each finding shows its evidence entries and their source crops.

FR-10 Insufficiency action

When R3 fires, compute the required earlier search window (from the earliest PR year, minus a configurable buffer, to the day before the current window opens) and render a pre-filled EC order block: SRO, village, survey numbers, date range.

FR-11 Correction

Any typed field is editable. On save: persist `{entry_id, field, old, new, actor, timestamp}`, re-run FR-6 → FR-9, and visibly update affected findings. Corrections survive restart and are attributed.

FR-12 Case persistence

Cases, EC documents, entries, parties, edges, findings, corrections persist in SQLite. Reopening a case restores full state including which findings were reviewed.

FR-13 Export

Generate a scrutiny report (HTML/PDF) containing property identifiers, search period, the chain, all findings with evidence, an explicit *"what this search does not cover"* section, the rulebook version, and a correction log.

FR-14 Staleness

Display EC issue date and days elapsed; flag beyond a configurable threshold.

9 Non-Functional Requirements

DIMENSION	REQUIREMENT	BASIS
Performance	≤ 90 s from upload to findings for a ≤ 10 -page EC	Measured: 12 pages ≈ 60 s
Throughput	Respect 10 req/min on Document Digitization; queue and back off	Documented hard limit, all plans
Reliability	Any Sarvam call retries with exponential backoff; a failed chunk degrades that page, never the case	Async job API can fail mid-poll
Correctness	100% of legally sensitive verdicts produced by deterministic code, 0% by model output	Non-negotiable design rule
Scalability	Stateless workers, job queue; per-case data isolated	Post-hackathon
Security	API key server-side only, never in the browser; uploads scoped to the owning advocate	.env already gitignored
Privacy	ECs are public records but name real parties; no third-party analytics, no sending documents anywhere but Sarvam; deletable case files	Professional obligation
Cost	$< ₹5$ per EC ($₹0.5/\text{page}$ + negligible tokens)	Measured: ₹6 for 12 pages
Auditability	Raw Sarvam responses retained immutably alongside every derived record	Provenance is the product

10 AI Architecture

10.1 Division of labour (the central design rule)

THE RULE

Models read. Code decides.

Sarvam Vision converts pixels to structured text. 105B converts structured text to typed records. Both are *transcription* roles. Every judgment a lawyer would be liable for — is this encumbrance live, is the chain broken, is this search sufficient — is executed by deterministic Python over typed data, is unit-testable, and is explainable by citing the rule and its evidence. This is a product-integrity decision first and a rubric decision second, and it is what separates TitleChain from "upload the PDF to a chatbot."

10.2 Models

STAGE	MODEL	WHY	CONFIG
Digitisation	Sarvam Vision (Document Digitization)	Only model verified at 98.6% on real Tamil EC tables with coordinates	language=ta-IN, output_format=md
Entry typing	sarvam-105b	128K context holds a full EC's table HTML; strict json_schema guarantees parseable output	temperature=0, seed fixed, response_format=json_schema strict
Name normalisation	Transliteration API	Deterministic, purpose-built for Indic↔Roman	ta-IN→en-IN

sarvam-30b is the fallback for typing if 105B rate-limits (40 req/min Starter on both).

10.3 Prompting strategy

A single system prompt for typing, with these constraints:

- Role: **transcriber, not analyst**. Copy values; do not interpret, summarise, translate, or resolve contradictions.
- Preserve Tamil text verbatim in *_native fields; never translate names.
- Unreadable or absent → null. **Never infer a plausible value.** (This is what makes the completeness validator meaningful.)
- Emit one object per numbered EC entry; sub-rows (consideration, market value, PR, remarks, schedules) attach to their parent entry.
- Copy PR numbers and remark text exactly — they are the graph edges.

Temperature 0 and a fixed seed, so a re-run is reproducible and a disagreement between two runs is a real signal.

10.4 Extraction schema (strict)

```
{
  "ec_header": {
    "sro": "string", "village": "string",
    "survey_details": ["string"],
    "search_period_start": "string|null", "search_period_end": "string|null",
    "issue_date": "string|null", "declared_entry_count": "integer|null"
  },
  "entries": [{
    "sr_no": "integer",
    "doc_no": "string|null", "doc_year": "integer|null",
    "date_execution": "string|null", "date_presentation": "string|null",
    "date_registration": "string|null",
    "nature": "string|null",
    "executants": [{"name_native": "string", "role_marker": "string|null"}],
    "claimants": [{"name_native": "string", "role_marker": "string|null"}],
    "volume_page": "string|null",
    "consideration_value": "string|null", "market_value": "string|null",
    "pr_numbers": [{"doc_no": "string", "year": "integer"}],
    "remarks": "string|null",
    "schedules": [{
      "property_type": "string|null", "extent": "string|null",
      "village_street": "string|null", "survey_nos": ["string"],
      "door_no": "string|null", "boundaries_native": "string|null"
    }]
  }
}]
```



```
"source": {"page_num": "integer", "block_id": "string"}
}]
}
```

declared_entry_count is captured deliberately: the EC states its own entry count (பதிவுகளின் எண்ணிக்கை: 2), giving us a **free ground-truth checksum** against the number of entries we extracted. Cheap, deterministic, and catches whole-entry loss.

10.5 The rulebook (deterministic, ours)

ID	RULE	SEVERITY	FIRES ON OUR REAL DATA
R1	CANCELLED_INSTRUMENT — entry X's remarks name a cancelling document Y present in this EC → X extinguished; must not be reported as a live encumbrance	Material	Yes — EC-A
R2	LIVE_ENCUMBRANCE — nature ∈ {mortgage, simple mortgage, deposit of title deeds, charge, lien} with no matching discharge/release → OPEN	Blocking	—
R3	WINDOW_INSUFFICIENT — any PR year < search-period start year → chain continues outside this certificate	Blocking	Yes — both ECs
R4	DANGLING_PARENT — PR pointer with no corresponding entry in this EC → parent document unexamined	Material	Yes — both ECs
R5	CHAIN_BREAK — claimant cluster of entry <i>n</i> never appears as executant in a later transfer	Blocking	—
R6	SURVEY_DRIFT — survey numbers vary across entries without a subdivision/partition deed	Material	—
R7	EXTENT_MISMATCH — schedule extent changes without a partition instrument	Material	—
R8	STALE_EC — issue date older than threshold	Informational	Yes — EC-A issued 2023
R9	STRUCTURAL_GAP — a required field is null (completeness validator)	Material	Yes — EC-A entry 2 dates
R10	ENTRY_COUNT_MISMATCH — extracted entries ≠ declared_entry_count	Blocking	—

Six of ten rules fire on documents we already hold. The demo is not staged; it is what the data does.

10.6 The uncertainty net

Three independent signals, because \$0.2 proved one is insufficient:

1. **Block confidence** (from Sarvam) — triages whole regions. Useful, but blind to cell omissions.
2. **Schema completeness (R9)** — a required field is null. This is the signal that would have caught our one real failure inside a 0.96-confidence block.
3. **Entry-count checksum (R10)** — the document's own declared count vs. ours.

Optional fourth (F13): two extractions at different seeds; fields that disagree are surfaced as **competing readings** shown side by side against the source crop — the rubric's L5 wording, achieved literally.

10.7 Memory and context strategy

Three layers, matching the L5 definition:

- **Current task** — the open case: which EC, which entries, which findings reviewed, which corrections pending.
- **Relevant history** — prior corrections by this advocate, the party-name clusters accumulated across their cases, previously processed ECs for the same property.
- **Business rules** — the versioned rulebook. Every finding records the rulebook version that produced it, so a stale finding is distinguishable from a current one.

Corrections propagate: editing a name re-runs clustering, graph assembly and every rule, and visibly updates findings. Access is scoped to the owning advocate (F11).

10.8 Failure handling

FAILURE	RESPONSE
Digitisation job fails	Retry once; then mark that chunk UNREAD and show the raw page — never silently drop pages
105B returns unparseable JSON	Strict schema makes this unlikely; on failure retry once, then fall back to 30B, then flag the entry for manual typing

FAILURE	RESPONSE
Field unreadable	null → R9 fires → advocate reviews against the source crop
Rate limit (10/min)	Queue with backoff; UI shows honest queue position
Total Sarvam outage during demo	Pre-cached outputs for the three demo ECs (\$15.5)

11 System Architecture

FRONTEND – React + Vite (single page)

Upload · Chain graph · Findings list · Source-crop viewer · Inline correction · Report export

↓ REST/JSON

BACKEND – Python FastAPI (sarvamai SDK already proven working)

`ingest.py` – chunk >10p, validate, store original
`digitise.py` – Sarvam Document Digitization
`extract.py` – sarvam-105b, json_schema strict
`names.py` – Transliteration ta-IN→en-IN
`graph.py` – **OURS** – nodes, PR/cancel/succession edges
`rulebook.py` – **OURS** – R1..R10, pure functions, unit-tested
`crops.py` – **OURS** – bbox → cropped PNG from page raster
`report.py` – **OURS** – scrutiny report generation

↓

STORAGE – SQLite (file-backed, survives restart)

cases · ec_documents · entries · parties · name_clusters · edges · findings · corrections · rulebook_versions
 + filesystem: original uploads, raw Sarvam JSON, page rasters

External: Sarvam APIs only. No third-party services. Key server-side.

WHY SQLITE OVER SUPABASE

Zero setup cost, survives restart (which is what Memory is actually scored on), and no network dependency during the demo. Supabase is permitted by the rules but buys us nothing here.

Data model (core tables)

- cases(id, advocate_id, property_key, created_at, rulebook_version)
- ec_documents(id, case_id, sro, village, survey_nos, search_start, search_end, issue_date, declared_entry_count, page_count, file_path)
- entries(id, ec_id, sr_no, doc_no, doc_year, date_execution, date_presentation, date_registration, nature, volume_page, consideration_value, market_value, remarks, page_num, block_id, block_confidence)
- parties(id, entry_id, role, name_native, name_roman, role_marker, cluster_id)
- edges(id, case_id, from_entry, to_doc_no, to_doc_year, edge_type, resolved_entry_id NULL)
- findings(id, case_id, rule_id, severity, message, evidence_entry_ids, rulebook_version, status)
- corrections(id, entry_id, field, old_value, new_value, actor, created_at)

12 Success Metrics

Product

- Time from upload to findings (target ≤ 90 s for ≤ 10 pages)
- % of ECs processed without manual intervention (target $\geq 85\%$)
- Findings reviewed per case / corrections per case (falling over time = learning)

Business

- Advocate hours saved per file (target ≥ 2 h; A6)
- Cost per EC (target $< ₹5$)
- Files per advocate per month (capacity increase)
- Retention: cases per advocate in month 2 vs month 1

AI quality

- **Field-anchor fidelity** — the $\$0.2$ metric, re-run on every new EC (current: 98.6%)
- Entry-count checksum pass rate (R10)
- Structural-gap rate (R9 fires per 100 entries)
- Correction rate per field type — identifies which fields the pipeline is weakest on
- Rule precision: % of flagged findings the advocate confirms rather than dismisses

User experience

- Can a first-time advocate complete a case without help? (binary, tested on the day)
- Time to locate the source evidence for a finding (target < 5 s via crop)
- % of findings where the advocate clicks through to source (engagement with provenance = trust)

13 Competitive Landscape

	WHAT THEY DO	WHAT WE CHANGE
Manual scrutiny	Human reads the table	We make the mechanical failures structurally impossible to miss, and leave judgment to the human
Record retrieval platforms	Get you the EC faster	We start where they stop. Access ≠ interpretation. Complementary, not competitive.
Generic LLM upload	Narrates a summary	We never let a model decide an encumbrance. Deterministic rules + provenance + an honest "this evidence is insufficient" — which a summariser structurally cannot produce, because it does not know what it was not given.
IDP/OCR suites	Extract fields	They return fields; we return a graph, a rulebook verdict, and a next action
Human vetting vendors	Outsource the reading	Same failure modes at lower cost; no accumulating asset

THE STRUCTURAL DIFFERENCE, STATED PLAINLY

Everyone else treats the EC as a document to be read. We treat it as a partially-observed graph, and the most valuable thing we output is not what the document says – it is what the document cannot tell you, and exactly which document to order next. No competitor is positioned to say that, because saying it requires modelling the chain rather than the page.

14 Build Plan

14.0 Tonight (pre-event – research and validation only, NOT product code)

COMPLIANCE

Buildathon rules disqualify "code written off the floor." Everything below is capability validation and test-data sourcing, which the rules permit as research. **All product code is written on the floor tomorrow.** `run_di.py` and `check_ec_integrity.py` are validation harnesses and will not be carried into the product; the pipeline is rebuilt from zero on the day. This will be declared in the submission notes as a borderline starting point – *hiding origin is auto-disqualification, so we declare it.*

- **Stage-2 dry run** – feed a digitised EC's table HTML to `sarvam-105b` with the §10.4 schema; confirm typed rows come back clean. **This is the last untested load-bearing dependency.**
- **Source 3+ ECs from different SROs/districts** – kills the Pollachi-only formatting risk.
- **True phone-photo test** – print one EC, photograph it at an angle under room light, run it. This is the honest version of the degraded-input class.
- Confirm API credit balance.

14.1 • Hour 1

10:30-11:30

Prove the hardest dependency end to end

Goal: one ugly, hardcoded, complete pass. EC PDF in → typed entries → one rule firing → printed output. No UI, no database, no styling.

Tasks: FastAPI skeleton · digitise call · table-block isolation · 105B typing with strict schema · R3 (window insufficiency) hardcoded · print findings to console.

Acceptance: on EC-A, the console prints the two entries and correctly states that five PR documents fall outside the search window.

Deliverable: working `main.py`, committed.

Fallback: run against pre-cached digitisation output and build stage 2 first – digitisation is already proven, so it's the safe thing to stub.

Checkpoint: idea committed by 11:30 ✓ · running by 12:15 ✓

14.2 • MVP

11:30-14:00

The product

Graph assembly (PR + cancellation + succession edges) · full rulebook R1–R10 · SQLite persistence · name clustering via transliteration · bbox crop generation · minimal React UI (upload, graph, findings, crop viewer).

Acceptance: EC-A and EC-B both produce correct graphs and findings; state survives a server restart; every finding renders its source crop.

Deliverable: demo-able product on two real ECs.

Fallback: cut the graph *visualisation* to a nested list; keep the graph *logic*. Cut F13.

14.3 • Beta

14:00-15:30

The scoring surface

Correction UI with propagation (F11 memory proof) · pre-filled next-EC-order block (F5 delight beat) · report export (F10 artifact) · advocate login + case scoping · staleness clock.

Acceptance: correcting a party name visibly re-derives the chain and updates findings; correction survives restart and is attributed; report exports with the "what this search does not cover" section.

Fallback: cut login (F11) and staleness (F12) – keep correction propagation, the actual Memory evidence.

14.4 • Final

15:30-16:30

Reserved – do not build

No new features after 15:30. This block is: three repeated runs on unseen ECs · state reset script · fallback inputs staged · pre-cached demo outputs · submission assets · **two timed rehearsals** · borderline-starting-point declaration written.

14.5 Production vision (post-event)

Multi-EC merge per property → sale-deed cross-verification → district rulebook profiles → firm-level name graph → bank portfolio scanning → adjacent states.

15 Demo Strategy

15.1 The 90-second script

T	ACTION	WHAT IT PROVES
0:00–0:10	"Every property loan in Tamil Nadu needs a 13-year title search. This is the certificate it runs on." <i>Show the raw Tamil EC.</i>	Hard, real, India-specific input
0:10–0:25	Upload EC-A. Processing visible. Entries appear as typed rows with Tamil names intact.	Document Intelligence
0:25–0:40	"Read linearly, this lease looks like a live encumbrance." Show the graph: 2520/2019 —cancelled-by→ 8756/2020. "It was cancelled in 2020. It is not an encumbrance."	Creativity — the graph reframing
0:40–1:00	"But here is what this certificate can't tell you." R3 fires: five parent documents from 2005–2011 sit outside the 2018–2023 window. "This EC cannot support a 13-year search." Pre-filled order for the exact replacement certificate appears.	Delight — the anticipated pain point + next action
1:00–1:15	Click a finding → source crop highlights the exact region on the Tamil page. Click the flagged missing date → "we could not read this; here is where it is."	Document Intelligence (uncertainty made precise)
1:15–1:30	Correct a party name → chain re-derives → close the case → reopen → correction and findings intact. Export the report.	Memory & Context + JTBD (final artifact)

Held in reserve if asked: run EC-B live, unseen — proves repeatability without builder rescue (JTBD L5).

15.2 Demo inputs

- **Primary:** EC-A (`ec_samples/ec2_pacollege.pdf`) — the cancellation trap + window insufficiency. Real, public, unmodified.
- **Secondary (live, for repeatability):** EC-B (`ec_samples/ec3_rera.pdf`) — 6 pages, 11 schedules, gift deed to a town panchayat.
- **Third:** an EC from a different SRO, sourced tonight — proves we are not overfit to Pollachi.

15.3 Expected outputs

EC-A: 2 entries · 2 graph nodes · 1 CANCELLED_INSTRUMENT · 1 WINDOW_INSUFFICIENT (5 parents, 2005–2011) · 5 DANGLING_PARENT · 1 STRUCTURAL_GAP (entry 2 dates) · 1 STALE_EC · pre-filled order for 01-Jan-1993 → 31-Dec-2017, Pollachi SRO, Puliampatti, survey 95/2, 100/3A, 113/1B, 116/A1, 116/B1.

15.4 Backup inputs

The degraded-scan variant (`ec_samples/ec2_scan.zip`) — scored 100%, and doubles as a "works on scans too" beat if there is time. The nil-EC portal screenshot (`ec_samples/negative_nil-ec-portal-screenshot.pdf`) is retained as a negative case: the product should say "this is not an EC" rather than hallucinate entries.

15.5 Failure recovery

1. **Network/API down:** pre-cached digitisation outputs for all three ECs on disk; a `--offline` flag replays them through the real graph and rulebook. The reasoning layer — which is ours and is the actual product — still runs live.
2. **Slow processing:** start the upload while delivering the opening line; the 60 s measured runtime fits the script.
3. **Unexpected extraction error:** it becomes the demo. Show the structural-gap flag catching it and the source crop letting the advocate fix it in five seconds. *The product's thesis is that it exposes uncertainty rather than hiding it — a visible catch is on-message.*
4. **Laptop/projector failure:** exported PDF report of the EC-A case on a USB stick and in the browser.

16 Evidence Map

One piece of evidence per parameter. No evidence is reused – per the anti-double-counting rule.

FEATURE	JTBD	MEMORY	CREATIVITY	IMPACT	DELIGHT	DOC INTELLIGENCE
F1 Digitisation of Tamil EC tables						the 98.6% fidelity + source crops
F2 Entry typing	supporting		✓			
F3 Graph assembly from PR pointers			the reframing: document → graph			
F4 Deterministic rulebook	supporting		supporting		✓	
F5 Window insufficiency + pre-filled order			supporting		anticipates the unasked question, gives the next action	
F6 Source-crop provenance						supporting
F7 Completeness validator		✓				supporting
F8 Correction propagation + persistence		correction survives restart, re-derives graph, stays attributed				
F9 Name clustering	supporting ✓					
F10 Exported scrutiny report	the final usable artifact, produced end to end unaided					
F11 Login / case scoping		supporting		✓		
Business case §4.2				~40% of scrutiny effort, defended with stated assumptions		

Every mandatory feature maps to a parameter. Anything that maps to nothing is in the parking lot.

17 Risks

Technical

RISK	SEV	MITIGATION
Stage-2 typing produces malformed/incomplete rows	High	Strict json_schema; verify tonight (\$14.0); 30B fallback; R9/R10 catch what slips
EC formats vary across SROs	High	Source 3 districts tonight; extraction is layout-agnostic (LLM over HTML, not fixed coordinates)
Real phone photos degrade worse than our synthetic scan	Medium	Test tonight; digital PDFs are the dominant real-world case (ECs are downloaded, not photographed)
Confidence score can't catch cell omissions	Confirmed real	Already mitigated by design: completeness validator + entry-count checksum
10 req/min rate limit	Low	Measured headroom is ample; queue with backoff

Product

RISK	SEV	MITIGATION
Advocates distrust automated findings	Medium	Deterministic rules + source crops + mandatory human sign-off; we evidence, never opine
Liability if a flag is wrong	Medium	Explicit scope disclaimer; the report states what was <i>not</i> covered
Impact assumptions don't survive scrutiny	Medium	Assumptions labelled and sensitivity-tested (\$4.1–4.2)

Demo

RISK	SEV	MITIGATION
Live API failure	Medium	--offline cached replay; our layer still runs live
Demo runs long	Medium	Two timed rehearsals in the reserved block; 90 s script with a hard cut list
Judges see "OCR + summary"	High	Lead with the insufficiency verdict, not the extraction. Extraction is 15 seconds of a 90-second demo.

SARVAM LIMITATIONS (KNOWN, DESIGNED AROUND)

10-page/job cap (chunk) · 10 req/min (queue) · async-only, no realtime vision (fits a desk workflow) · block confidence ≠ cell confidence (own validator) · no handwriting guarantee via API (out of scope by design) · no Modi/Kaithi/Grantha (out of scope by design).

Scope

RISK	SEV	MITIGATION
Feature creep during build	High	Parking lot in §7; no new features after 15:30; every feature must name a rubric parameter
Over-investment in graph visualisation	Medium	Nested list is an acceptable fallback; the logic is what scores
Borderline starting point	High	Rebuild all product code on the floor; declare pre-event validation in submission notes

18 Why This Wins

"This certificate cannot support the search it was ordered for, and here is the exact one to order instead."

Judges will remember it because of one sentence they did not expect. Every other document product at the event will extract, summarise, or translate. This one tells the user what their evidence does not cover — a conclusion only reachable by modelling the chain rather than the page, and one that is immediately, obviously correct once shown.

It scores across every dimension without padding. Document Intelligence is proven at 98.6% on real Tamil government tables with source-region traceability. JTBD produces a genuine professional artifact end to end. Memory is demonstrated by a correction that survives a restart and re-derives the graph, not by remembering a name. Creativity is structural — a document reframed as a partially-observed graph, which then makes the insufficiency check possible. Impact is a defended ~40% reduction in scrutiny effort with assumptions on the table. Delight is answering the question the user was afraid to ask. Nothing is decorative; the voice and dubbing layers are deliberately absent because the job does not need them.

It is hard to replicate on the day. Another team would need to find TN ECs, know that the PR Number field is a parent pointer, recognise that PR years routinely precede the search window, and discover — as we did, empirically — that the API's confidence score cannot catch a cell dropped from a high-confidence table. That last one is a trap most teams will walk into and ship without noticing.

It has a real path to a company. The reading problem is solved and commoditising; the *rulebook, the corrections and the name graph are not*, and they compound with every file. The wedge is one narrow certificate in one state, which is exactly how a title-graph company should start — the same logic extends to Karnataka, Telangana and Andhra Pradesh, where equivalent instruments carry the same parent pointers. The endgame is a verified title graph for Indian property, assembled from the documents transactions already produce.

A Appendix – Verified Capability Reference

PRIMITIVE	TIER	HARD LIMITS	OUTPUT GRANULARITY
Document Digitization (Sarvam Vision)	Explicit + empirically verified	10 pages/job, 200 MB, 10 req/min, ₹0.5/page, async	MD/HTML + per-page JSON: block_id, coordinates {x1, y1, x2, y2}, layout_tag, confidence, reading_order, text (tables as HTML). 3509×2480 raster.
sarvam-105b	Explicit	128K ctx, 40 req/min Starter, ₹4/₹16 per 1M tok	Text; strict json_schema; tool calling; no image input
Transliteration	Explicit	ta - IN supported both directions	Text + detected source

Assumed capabilities we are NOT depending on: handwriting via API · dashboard-only field extraction with confidence · "Vision real-time" · voice cloning/dubbing APIs · wiki_grounding (discontinued for 30B/105B).

B Appendix – Repository

PRD.md	this document
RUBRIC.md	event handbook and scoring ladders
IDEA_DISCOVERY_PROMPT.md	capability-first methodology that produced this idea
config.py, check_key.py	API key loading (validation harness)
run_di.py	digitisation harness (validation only – NOT product code)
check_ec_integrity.py	anchor-fidelity checker (validation only)
ec_samples/	real EC test inputs + degraded variants + negative case
output/	cached digitisation results (demo fallback)